

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	K Abishek	Reg. No.:	192521120
Section / Batch:	IT	Date:	23/07/26

Instructions to Students

- Answer the following question. It carries 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big- Θ , Big- Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I K ABISHEK/192521120(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K ABISHEK

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Smart Classroom Attendance System

Aim

To implement a smart classroom attendance system that scans each student's ID exactly once and marks their attendance. The time and space complexity of the implemented algorithm are analyzed.

Problem Statement

Develop a program that accepts the number of students and scans each student's ID exactly once. After scanning, the system marks the student as present. Analyze the time complexity and space complexity of the implemented program.

Algorithm

1. Start the program.
2. Read the number of students n .
3. Initialize a loop counter.
4. Repeat the following steps for each student from 1 to n :
 - Read or scan the student's ID.
 - Mark the student as present.
5. Display the attendance status.
6. Stop the program.

Source Code

```
# Smart Classroom Attendance System

n = int(input("Enter the number of students: "))

for i in range(1, n + 1):
    student_id = input(f"Scan Student {i} ID: ")
    print(f"Student ID {student_id} - Attendance Marked Present")

print("Attendance process completed successfully.")
```

Sample Input

```
Enter the number of students: 3
Scan Student 1 ID: ST101
Scan Student 2 ID: ST102
Scan Student 3 ID: ST103
```

Sample Output

```
Student ID ST101 - Attendance Marked Present
Student ID ST102 - Attendance Marked Present
Student ID ST103 - Attendance Marked Present
Attendance process completed successfully.
```

Implementation Explanation

The program first accepts the total number of students n . A `for` loop is then used to scan the ID of every student exactly once. For each scanned ID, the program marks the student's attendance as present.

Since the loop executes exactly n times, the number of operations increases directly with the number of students.

The total number of operations can be represented as:

$$T(n) = n$$

Therefore, the time complexity of the implementation is:

$$\text{Time Complexity} = O(n)$$

This is known as **Linear Time Complexity**.

Growth of Number of Operations

Number of Students (n)	Number of ID Scans
10	10
100	100
1000	1000
n	n

The number of ID scans grows directly in proportion to the number of students.

Space Complexity Analysis

The program uses only a few variables:

- `n` – stores the number of students.
- `i` – loop counter.
- `student_id` – stores the currently scanned student ID.

The program does not create an additional data structure such as an array or list to store all student IDs. Therefore, the extra memory used by the algorithm does not increase with the number of students.

Hence:

$$\text{Auxiliary Space} = O(1)$$

Therefore:

$$\text{Space Complexity} = O(1)$$

Asymptotic Complexity

- Time Complexity: $O(n)$
- Space Complexity: $O(1)$

Result

The Smart Classroom Attendance System was successfully implemented using a loop that scans each student ID exactly once and marks attendance. The implementation requires **$O(n)$ time** because each student is processed once and **$O(1)$ auxiliary space** because only a constant number of variables are used.

Conclusion

The implemented Smart Classroom Attendance System efficiently processes student attendance by scanning every student ID exactly once. As the number of students increases, the execution time increases linearly. Since the program uses only a constant amount of additional memory, its space complexity remains constant. Thus, the implementation has **$O(n)$ time complexity and $O(1)$ space complexity**.